

Customer Churn Prediction in Mobile Telecommunications Using K-Means Clustering and Ensemble Predictive Modelling

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Abstract

Customer churn poses a substantial revenue risk in the highly competitive mobile telecommunications sector. This paper presents an end-to-end churn prediction and intervention framework applied to a dataset of over 15 million customer records from a major East African mobile network operator. The methodology combines K-means clustering for behavioural segmentation with gradient boosted decision trees for probabilistic churn prediction, achieving an AUC-ROC of 0.863. Key findings include identification of a churn-prone behavioural cluster with a 14.2% monthly churn rate and quantification of a 6.3% average monthly churn rate across the portfolio. A targeted SMS intervention informed by model predictions achieved a statistically significant 24% relative churn reduction across 1.2 million high-value customers.

Keywords: customer churn · K-means clustering · gradient boosting · A/B testing · mobile money analytics · telecommunications · East Africa

1. Introduction

Acquiring a new customer costs five to seven times more than retaining an existing one (Reichheld & Sasser, 1990). In the Kenyan mobile market — where Safaricom, Airtel, and Telkom compete for subscribers with mobile penetration exceeding 130% (CA Kenya, 2024) — churn management is a strategic priority with direct P&L implications.

This paper addresses three research questions: (1) What distinct behavioural clusters exist within a large mobile subscriber base? (2) Can a supervised ML model accurately identify subscribers at high risk of churning within a 30-day window? (3) Does a model-informed targeted SMS campaign measurably reduce churn?

2. Literature Review

2.1 Churn Prediction Methods

Early churn prediction relied on logistic regression applied to aggregate usage statistics (Mozier et al., 2000). Gradient boosting methods (Friedman, 2001) consistently outperform classical methods on telecom churn benchmarks (Coussement & Van den Poel, 2008). Deep learning LSTM approaches show promise but face interpretability constraints in regulated environments.

2.2 Mobile Money and Churn

Jack & Suri (2011) document the centrality of M-Pesa to daily financial life for millions of Kenyans, suggesting that mobile money transaction frequency serves as a strong churn predictor — customers deeply embedded in the M-Pesa ecosystem face higher switching costs.

3. Data and Methodology

3.1 Dataset

The dataset comprises 15,247,000 anonymised customer records over a 12-month observation window. Churn is defined as zero airtime and data usage for 30 consecutive days. The observed average monthly churn rate was 6.3%. The estimated average daily M-Pesa spend was KSh 340 (σ = KSh 218).

Feature Category	Variables
Voice Usage	Outgoing/incoming minutes (weekly), unique contacts
Data Usage	Bundle purchases, MB consumed, top-up frequency
Mobile Money	M-Pesa send/receive count, average daily spend
Recharge Behaviour	Top-up frequency, average value, days since last
Customer Profile	Tenure (months), SIM type, regional class

3.2 K-Means Clustering

Features were standardised to zero mean and unit variance. The optimal cluster count $K = 5$ was determined using the elbow method (WCSS) and silhouette score analysis.

3.3 Churn Prediction Model

An XGBoost gradient boosted classifier was trained on 70% of records (~10.7M) and validated on 30% (~4.6M). Class imbalance (6.3% positive rate) was corrected via `scale_pos_weight = 14.9`. Hyperparameters were tuned via 5-fold cross-validation: `n_estimators=400`, `max_depth=6`, `learning_rate=0.05`.

3.4 A/B Test Design

A randomised controlled trial was conducted among predicted high-risk, high-value customers. Treatment group received personalised SMS retention offers; control group received no intervention. Stratified by cluster and predicted churn decile; powered at 80% to detect 0.5pp reduction ($\alpha = 0.05$).

4. Results

4.1 Customer Segments

Cluster	Label	Size	Avg Spend (KSh)	Churn Rate
1	High-Value Loyalists	8.2%	3,240	1.8%
2	Data-First Millennials	22.1%	1,180	5.4%
3	Voice-Centric Seniors	18.7%	890	6.9%
4	Dormant / At-Risk	31.4%	210	14.2%
5	Mobile Money Power Users	19.6%	1,560	2.6%

Cluster 4 (Dormant/At-Risk) exhibited a churn rate of 14.2% — more than twice the portfolio average — characterised by declining recharge frequency, low M-Pesa engagement, and high data bundle abandonment.

4.2 Model Performance

Metric	Value
AUC-ROC	0.863
AUC-PR	0.641
F1-Score	0.592
Precision	0.611
Recall	0.574
Log-Loss	0.198

4.3 Feature Importance

The top predictors by XGBoost gain: (1) Days since last recharge (0.189); (2) Recency-weighted engagement score (0.164); (3) M-Pesa transaction count — 30 days (0.141); (4) Data bundle purchase frequency (0.118); (5) Outgoing call minutes trend (0.097). M-Pesa frequency ranks third, confirming that mobile money embeddedness is a powerful retention signal.

4.4 A/B Test Results

Group	Sample Size	30-Day Churn Rate	95% CI
Treatment	612,000	4.87%	[4.71%, 5.03%]
Control	588,000	6.41%	[6.23%, 6.59%]

The treatment group experienced a 1.54 percentage point reduction in 30-day churn (24% relative reduction), statistically significant at $\alpha = 0.001$ ($p < 0.0001$). Annualised revenue retained across 1.2M customers is estimated at KSh 739 million.

5. Discussion

The outsized predictive importance of M-Pesa transaction frequency supports the theory that mobile money creates switching costs beyond voice and data. Customers deeply embedded in the M-Pesa ecosystem face a higher cost of switching operators than a simple tariff comparison would suggest.

Churn peaks in January (8.1%) and July (7.8%) coincide with school fee cycles, enabling predictable pre-peak intervention timing in December and June.

6. Conclusion

This paper presents a comprehensive, production-validated churn prediction framework achieving $AUC = 0.863$ on 15M+ records with a 24% relative churn reduction in the treatment group. The M-Pesa embeddedness finding and seasonal patterns provide actionable guidance replicable across Sub-Saharan African telecom markets.

References

- Coussement, K., & Van den Poel, D. (2008). Churn prediction in subscription services. *Expert Systems with Applications*, 34(1), 313–327.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- Jack, W., & Suri, T. (2011). Mobile money: The economics of M-PESA. *NBER Working Paper No. 16721*.
- Mozer, M. C. et al. (2000). Predicting subscriber dissatisfaction. *IEEE Transactions on Neural Networks*, 11(3), 690–696.
- Reichheld, F. F., & Sasser, W. E. (1990). Zero defections: Quality comes to services. *Harvard Business Review*, 68(5), 105–111.